Crypto Tracker

1. Introduzione

1.1 Use case

Crypto Tracker is an application that allows users to monitor and consult information related to cryptocurrencies available on the market.

The application provides a graphical interface that allows users to:

- search available cryptocurrencies, applying optional filters
- add one or more cryptocurrencies to a favorites list for easier monitoring
- view detailed information about cryptocurrencies and analyze their historical price trends

The main use case is the following: the user selects one or more cryptocurrencies from a list, adds them to favorites, and then consults their details along with a chart representing the price history over a given time interval.

1.2 Structure

Crypto Tracker is structured into two parts:

- client: a JavaFX application that provides the graphical interface and manages interaction with the service
- server: a web service developed using Spring Boot, responsible for data management and interaction with an external service

Communication between client and server occurs via the HTTP protocol, with data exchanged in JSON format.

The server retrieves cryptocurrency data by querying an external web service available at: https://free-apis.github.io/ (Cryptocurrency section).

In particular, the CoinGecko API is used, whose official documentation is available at: https://docs.coingecko.com/

2. Application

2.1 Main screen

At application startup, a main screen is displayed containing four buttons:

- **initialize**: loads cryptocurrency data into the database, removing any existing data
- **search**: displays a table containing basic information about cryptocurrencies
- **favorites**: shows only the cryptocurrencies added to favorites

- **update**: updates the data stored in the database with the most recent values

2.2 Search

In this screen, a list is displayed containing the main information about the top 100 cryptocurrencies sorted by **market cap rank**, defined as the product between unit price and circulating supply. This is the main parameter used to evaluate the size and relevance of a cryptocurrency.

Initially, all cryptocurrencies stored in the database are shown; however, the user can apply filters based on:

- minimum price
- maximum price
- positive price variation in the last 24 hours

After setting the filters, the user must click the **“Apply”** button. By right-clicking on a row in the table, the user can add or remove the selected cryptocurrency from the favorites list.

Name	Market cap rank	Current price (€)	Price change percentage 24h (%)	★
Bitcoin	1	81803.0	-1.06466	false
Ethereum	2	2838.71	-1.19292	false
Tether	3	0.861329	-0.04589	false
BNB	4	800.94	-0.69477	false
XRP	5	1.77	-2.05842	false
Solana	6	122.97	-0.34674	false
USDC	7	0.861512	-0.02251	false
Lido Staked Ether	8	2842.2	-1.2927	false
TRON	9	0.264368	-0.31497	false
Dogecoin	10	0.118289	-3.62104	false
Figure Heloc	11	0.887365	-0.79629	false
Cardano	12	0.333974	-3.59755	false
Monero	13	612.6	0.29941	false

2.3 Favorites

In this screen, only the cryptocurrencies previously added to favorites are displayed.

The **"Remove all"** button allows clearing the entire favorites list.

By right-clicking on an element in the table, it is possible to remove it from favorites or access the details screen.

Name	Market cap rank	Current price	Price change percentage 24h
Stellar	26	0.192374	-3.43876
Dai	38	0.861568	-0.07269
Polkadot	44	1.81	-3.58252
Bitget Token	50	3.25	0.80098
Tether Gold	54	3961.84	-0.21389
KuCoin	74	9.88	-0.95495
Cosmos Hub	86	2.14	-0.28455
Filecoin	95	1.3	-2.19785

2.4 Details

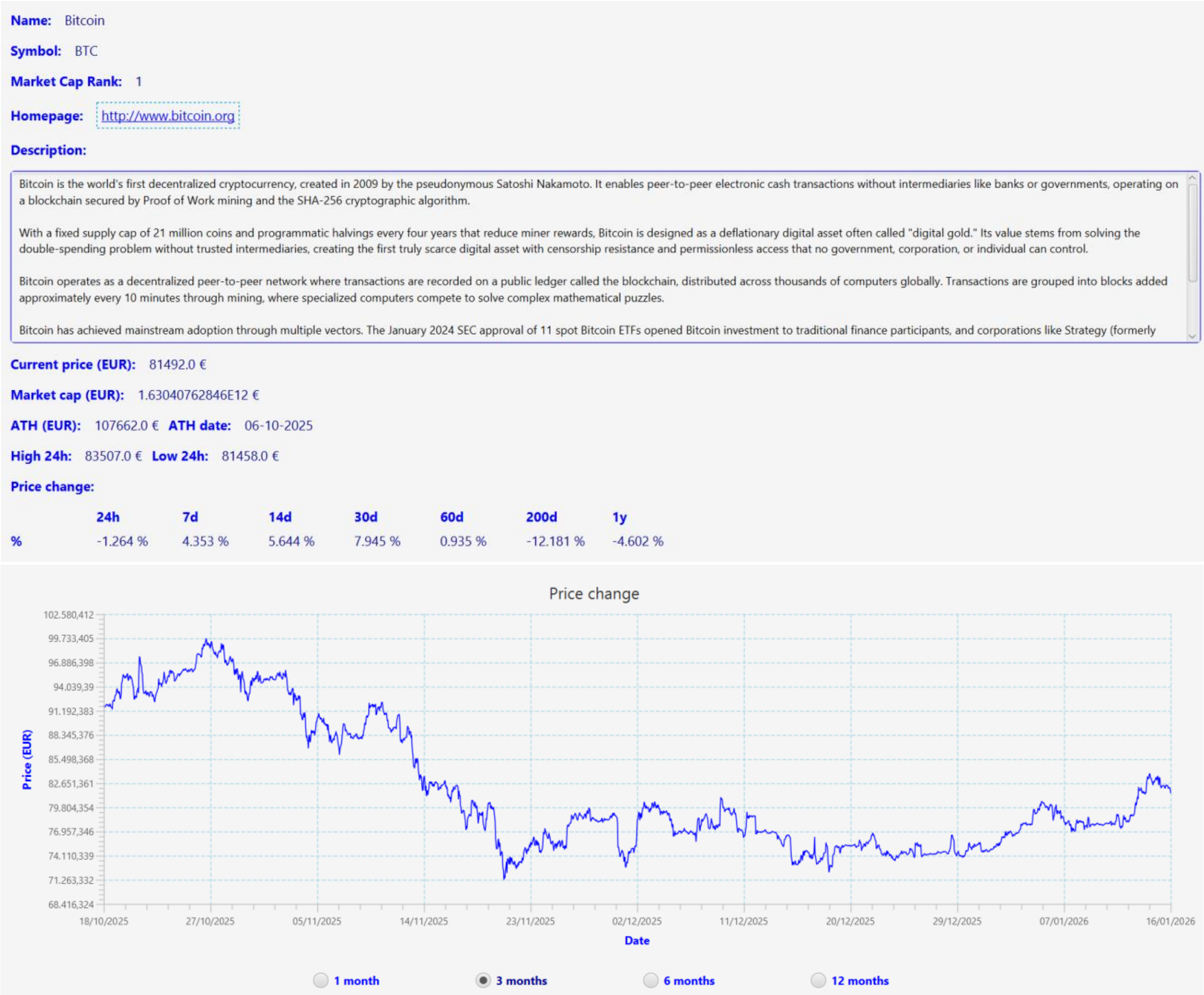
This screen contains additional information about the selected cryptocurrency, including:

- link to the official homepage
- a short description

- relevant market values

A chart representing the price trend over time is also displayed.

Initially, the chart shows data for the last three months, but the user can select different time intervals (last month, last six months, last year) using dedicated buttons.



2.5 Update

The CoinGecko API provides updated cryptocurrency data, but in the free version it imposes a strict limit on the number of requests that can be made within a certain time interval (**request rate**).

To reduce the number of requests to the external service, the server uses the database. Requests to CoinGecko are made only in the following cases:

- database initialization
- first request for cryptocurrency details
- first request for chart data

The retrieved data is stored in the database, and subsequent requests for the same information are handled by querying the database directly.

This approach reduces the load on the external service, but introduces the risk of using outdated data.

To solve this issue, an **“Update”** button is provided, which forces the update of all cryptocurrency data in the database by querying CoinGecko again.

Since this operation requires many consecutive requests, updates are performed with a 12-second interval between requests to avoid the **Too Many Requests** error.

The operation may take several minutes to complete.

3. Server

3.1 Database structure

At startup, the service creates a database named **“676518”** (if not already present) with the following tables:

- **crypto**: contains basic cryptocurrency information shown in the search screen, including name, market cap rank, current price, 24-hour percentage change, and favorite flag
- **crypto_details**: stores the JSON returned by CoinGecko for cryptocurrencies whose details have been requested
- **crypto_chart**: stores the JSON returned by CoinGecko for cryptocurrencies whose chart data has been requested

3.2 CoinGecko

Below are the CoinGecko API documentations used by the service:

- Initialization: https://docs.coingecko.com/reference/coins-markets
- Details: https://docs.coingecko.com/reference/coins-id
- Chart: https://docs.coingecko.com/reference/coins-id-market-chart

3.3 API list

The service exposes the following HTTP endpoints:

- **POST /inizializza**: sends an HTTP request to CoinGecko and initializes the “crypto” table
- **GET /search**: queries the database (applying user filters if present) and returns matching cryptocurrencies
- **PUT /{id}/favorite**: sets the “favorite” value of a cryptocurrency (true/false via query string)
- **GET /favorites**: returns only cryptocurrencies marked as favorites
- **PUT /clear**: sets the “favorite” field of all cryptocurrencies to false
- **PUT /update**: updates the database with the latest data (one request every 12 seconds)
- **GET /{id}/details**: retrieves detailed information about a cryptocurrency from CoinGecko
- **GET /{id}/chart**: retrieves chart data for the last “days” days (e.g., 30, 90, 180, 365)

3.4 Unit Test

The service includes a unit test implemented in the class **ServizioApplicationTest**, generated by Spring Boot.

The test verifies the correct behavior of the **GET /search** endpoint by:

- checking that the response has HTTP status code 200 (OK)
- verifying that the response content is encoded in JSON format